Maxwell's equations :-

$\nabla \cdot \bar{D} = \rho_v$ (gauss law) $\quad \nabla \cdot \bar{B} = 0$ (no free charge) $\quad \nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t}$ (faraday's law) $\quad \nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$ (ampere's law)

$\bar{D} = \varepsilon \bar{E} = \varepsilon_0 (1 + \chi_e) \bar{E} = \varepsilon_0 \varepsilon_r \bar{E}$

$\nabla = \hat{a}_x \frac{\partial}{\partial x} + \hat{a}_y \frac{\partial}{\partial y} + \hat{a}_z \frac{\partial}{\partial z}$

$\bar{B} = \mu \bar{H} = \mu_0 (1 + \chi_m) \bar{H} = \mu_0 \mu_r \bar{H}$

$\nabla \times \bar{E} = -\mu \frac{\partial \bar{H}}{\partial t} = -j\omega \mu \bar{H}$

$\nabla \times \bar{B} = \sigma \bar{E} + \epsilon \frac{\partial \bar{E}}{\partial t}$

$= \sigma \bar{E} + j\omega \epsilon \bar{E}$

$\boxed{\sigma, \epsilon, \mu \rightarrow RLC \Rightarrow \text{similar}}$

$\left.\begin{matrix} \sigma(x,y,z,t) \\ \epsilon(x,y,z,t) \\ \mu(x,y,z,t) \end{matrix}\right\}$ homogeneous medium $\Rightarrow$ constant

$\nabla \times (\nabla \times \bar{E}) = -j\omega \mu (\nabla \times \bar{H}) = -j\omega \mu \{\sigma \bar{E} + j\omega \epsilon \bar{E}\}$

$= \nabla (\nabla \cdot \bar{E}) - \nabla^2 \bar{E} = [-j\omega \mu \sigma + \omega^2 \epsilon] \bar{E}$

$\nabla \cdot \bar{D} = 0 \Rightarrow \nabla \cdot (\epsilon \bar{E}) = 0 \Rightarrow \nabla \cdot \bar{E} = 0$

$\Rightarrow \nabla^2 \bar{E} = [j\omega \mu \sigma - \omega^2 \epsilon] \bar{E}$

$\Rightarrow \boxed{\nabla^2 \bar{E} - \gamma^2 \bar{E} = 0} \quad \boxed{\nabla^2 \bar{H} - \gamma^2 \bar{H} = 0}$

$\boxed{\gamma^2 = j\omega \mu \sigma - \omega^2 \epsilon}$

$\gamma = \alpha + j\beta$

$\alpha = \boxed{\omega \sqrt{\frac{\mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1\right]}}$

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\epsilon}\right)^2}-1\right]}$$

$$\beta = \omega \sqrt{\frac{\mu\epsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\epsilon}\right)^2}+1\right]}$$

$v = 1/\sqrt{\mu\epsilon}$ for $\sigma = 0$ $\quad (v = \omega/\beta)$

Assume $\bar{E} = E_{xs}(z)\,\hat{a}_x$

$\Rightarrow \nabla^2 E - \gamma^2 E = 0 \Rightarrow \frac{d^2 E_{xs}(z)}{dz^2} - \gamma^2 E_{xs}(z) = 0$

$\Rightarrow E_{x,s}(z) = E_0^+ e^{-\gamma z} + E_0^- e^{-\gamma z}$ ($E_0^- = 0$ we assume)

$E_{x,s}(z) = (E_0 e^{-\alpha z}) e^{-j\beta z}$

$$\vec{E}(z,t) = \hat{a}_x E_0 e^{-\alpha z} e^{j(\omega t - \beta z)}$$

$$\vec{E}(z,t) = E_0 e^{-\alpha z} \cos(\omega t - \beta z)\,\hat{a}_x$$

Assumptions → only x-component of electric field
→ isotropic medium

$\nabla \times \bar{E}_s = -j\omega\mu\bar{H} \rightarrow \hat{a}_y \frac{d}{dz} E_{xs} = -j\omega\mu\bar{H}$

$\Rightarrow \bar{H}_{ys} = \left(\frac{\alpha + j\beta}{j\omega\mu}\right) E_0 e^{-\alpha z} e^{-j\beta z}$

$H_0 = \frac{E_0(\alpha + j\beta)}{(j\omega\mu)}$

$\bar{H}_{ys} = H_0 e^{-\alpha z} e^{-j\beta z}$

$\vec{H}(z,t) = (H_0 e^{-\alpha z}) e^{j(\omega t - \beta z)}$

$$H_0 = \sqrt{\frac{\sigma + j\omega\epsilon}{j\omega\mu}}\, E_0 \qquad \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \frac{E_0}{H_0}$$

$1/\eta$

E_0, H_0 will not be in phase

(Analogos to transmission line $R=0$; $L=\mu$; $G=\sigma$; $C=\epsilon$)

$$\eta = |\eta| e^{j\theta_\eta} \Rightarrow |\eta| = \frac{\sqrt{\mu/\epsilon}}{\left[1+(\sigma/\omega\epsilon)^2\right]^{1/4}} \; ; \; \tan(2\theta_\eta) = \frac{\sigma}{\omega\epsilon} \Rightarrow \theta_\eta = \frac{1}{2}\tan^{-1}\left(\frac{\sigma}{\omega\epsilon}\right)$$

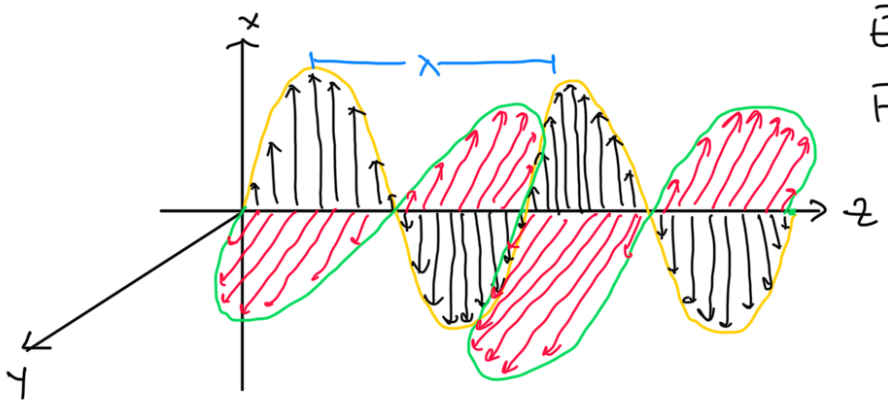
$$v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{\frac{\mu\epsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\epsilon}\right)^2}+1\right]}}$$

(phasefront is plane)

plane wave in free space $\Rightarrow \sigma=0$; $\epsilon=\epsilon_0$; $\mu=\mu_0$

$$\Rightarrow \alpha = 0 \; ; \; \beta = \omega\sqrt{\mu_0\epsilon_0} \Rightarrow c = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu_0\epsilon_0}} \approx 3\times10^8$$

$$\boxed{k_0 = \beta = \frac{2\pi}{\lambda_0}} \qquad \eta_0 = \sqrt{\mu_0/\epsilon_0} = 377 \Rightarrow \theta_{\eta_0} = 0$$

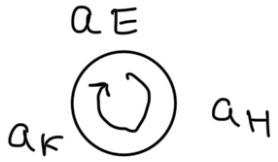


$$\bar{E}(z,t) = \hat{a}_x E_0 \cos(\omega t - kz)$$

$$\bar{H}(z,t) = \hat{a}_y H_0 \cos(\omega t - kz)$$

$$E_0/H_0 = \eta_0$$

$$\bar{\nabla}\times\bar{E} = -\mu\frac{\partial\vec{H}}{\partial t} \qquad \hat{a}_E \times \hat{a}_H = \hat{a}_R$$



sometimes E_0, H_0 aren't constant and depend on x or y). In that case

$$\bar{E}(x,y,z,t) = \hat{a}_x E_0(x,y)\cos(\omega t - kz)$$

$$\bar{H}(x,y,z,t) = \hat{a_y}\, H_0(x,y)\cos(\omega t - kz)$$

Conductive medium → $\sigma \to \infty$; ϵ ; μ

$$\alpha = \sqrt{\frac{\omega\sigma\mu}{2}} = \sqrt{\pi f \sigma \mu} \qquad f \to \text{linear frequency}$$

$$\beta = \sqrt{\pi f \sigma \mu} \qquad (\alpha = \beta) \neq 0 \Rightarrow \text{lossy}$$

$$z = \delta = 1/\alpha \Rightarrow \bar{E}(z,t) = E_0 e^{-\alpha z} e^{j(\omega t - \beta z)}$$

$$E_0 e^{-\alpha z} = \frac{E_0}{e} \quad \left(@\ \delta = 1/\alpha = 1/\sqrt{\pi f \sigma \mu}\right)$$

↓ skin depth

{higher frequency lower skin depth ?

Polarization

$$\vec{E}(z,t) = \hat{a_x} E_0 e^{j(\omega t - \beta z)} \qquad \gamma = \alpha + j\beta \quad \alpha = 0$$

$$\vec{H}(z,t) = \hat{a_y}\left(\frac{E_0}{\eta}\right) e^{j(\omega t - \beta z)} \qquad \frac{E_0}{H_0} = \eta = \sqrt{\frac{\mu}{\epsilon}}$$

$$\vec{E}(z,t) = \hat{a_x} E_{0x} e^{j(\omega t - \beta z)} + \hat{a_y} E_{0y} e^{j(\omega t - \beta z)}$$

linearly polarized:

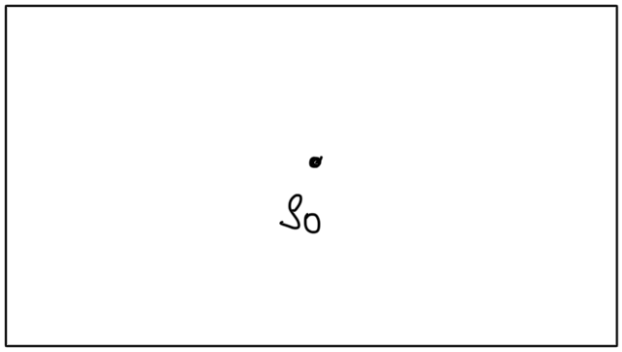
a_x, a_y component should be in phase

$$\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$$

$$\nabla\cdot(\nabla\times\bar{H}) = \nabla\cdot\bar{J} + \nabla\cdot\left(\frac{\partial \bar{D}}{\partial t}\right)$$

$$\Rightarrow 0 = \nabla\cdot\bar{J} + \frac{\partial \rho}{\partial t} \quad \rightarrow \text{continuity equation}$$

Question:-



at $t=0$ at some point charge ρ_0 is injected (initially neutral)

to find $\rho(t)$!

$$\nabla\cdot J + \frac{\partial \rho}{\partial t} = 0 \qquad J = \sigma E$$

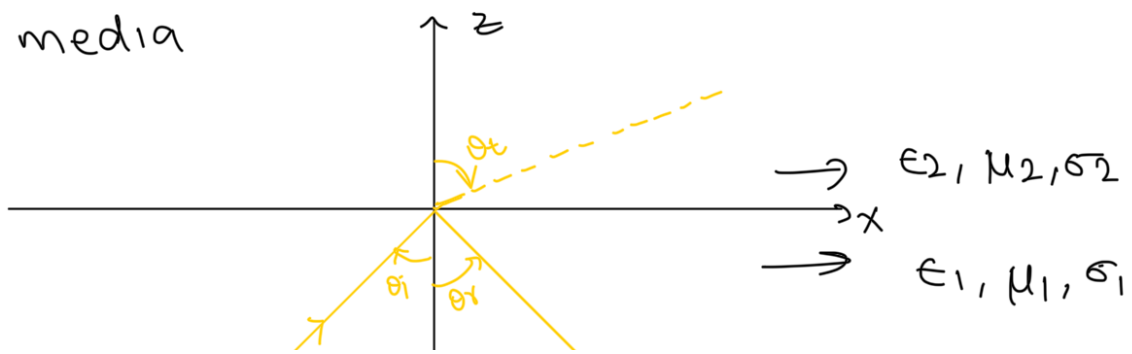
$$\frac{\sigma}{\varepsilon}\nabla\cdot(\varepsilon\bar{E}) + \frac{\partial \rho}{\partial t} = 0$$

$$\Rightarrow \frac{\sigma}{\varepsilon}(\nabla\cdot\bar{D}) + \frac{\partial \rho}{\partial t} = 0 \Rightarrow \rho(\sigma/\varepsilon) + \frac{\partial \rho}{\partial t} = 0$$

$$\Rightarrow \frac{d\rho}{\rho} = \left(-\frac{\sigma}{\varepsilon}\right)dt \Rightarrow \rho(t) = \rho_0 e^{-(\sigma/\varepsilon t)}$$

at $t\to\infty$; charge goes to surface and then disputes itself.

two media

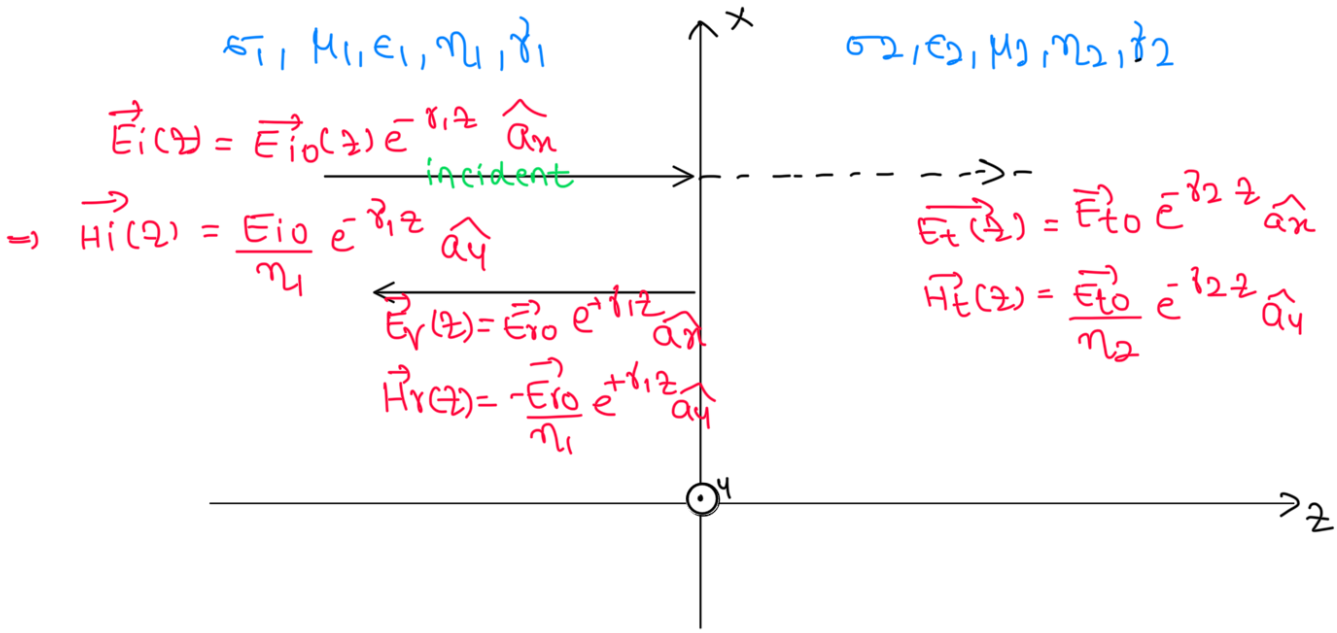


$\theta_i = \theta_r$
Snell's law } experimental $(\mu_1 = \mu_2)$

Boundary Conditions : $E_1^{\parallel} = E_2^{\parallel}$ $H_1^{\parallel} = H_2^{\parallel}$

$D_1^{\perp} = D_2^{\perp}$ $H_1^{\perp} = H_2^{\perp}$

Reflection of plane waves at Normal Incidence to a 2nd medium interface



medium 1: at $z=0 \Rightarrow E_1 = E_{i0} + E_{r0}$

medium 2: at $z=0 \Rightarrow E_1 = E_{t0}$

$\Rightarrow$ $\boxed{E_{i0} + E_{r0} = E_{t0}}$

also; $H_1 = \frac{1}{\eta_1}[E_{i0} - E_{r0}]$ $H_2 = \frac{E_{t0}}{\eta_2}$

$\Rightarrow$ $\boxed{\frac{1}{\eta_1}[E_{i0} - E_{r0}] = \frac{E_{t0}}{\eta_2}}$

η_1 η_2

$$E_{ro} = \left\{\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}\right\} E_{io}$$

$$E_{to} = \left\{\frac{2\eta_2}{\eta_2 + \eta_1}\right\} E_{io}$$

$$\Gamma = \frac{E_{ro}}{E_{io}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\tau = \frac{E_{to}}{E_{io}} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

$$\tau = 1 + \Gamma$$

Fresnel's Equations

$$0 \le |\Gamma| \le 1$$

If $\sigma_1 = \sigma_2 = 0$

$\mu_1 = \mu_2 = \mu_0$

let $\sqrt{\mu_0/\epsilon_0} = \eta_0$

$$\eta_1 = \sqrt{\frac{\mu_1}{\epsilon_1}} = \frac{\eta_0}{n_1} \qquad \eta_2 = \sqrt{\frac{\mu_2}{\epsilon_2}} = \frac{\eta_0}{n_2} \qquad n_i = \sqrt{\epsilon_{ri}}$$

$$\Rightarrow \Gamma = \frac{n_1 - n_2}{n_1 + n_2} \qquad \tau = \frac{2n_1}{n_1 + n_2}$$

$$\Gamma_{11} = \frac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i} \qquad \tau_{11} = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_i \cos\theta_i}$$

$\Gamma_{11} = 0$ for $\theta_i = \theta_{B11}$ (Brewster angle) $\qquad 1 + \Gamma_{11} = \tau_{11}$

$$\sin\theta_{B11} = \sqrt{\frac{1 - \frac{\mu_2 \epsilon_1}{\mu_1 \epsilon_2}}{1 - \left(\frac{\epsilon_1}{\epsilon_2}\right)^2}}$$

for $\mu_1 = \mu_2 = \mu_0 \Rightarrow \sin\theta_{B11} = \sqrt{\frac{\epsilon_2}{\epsilon_1 + \epsilon_2}} = \frac{n_2}{n_1 + n_2}$

$$\tan\theta_{B11} = \frac{n_2}{n_1}$$

$$\Gamma_\perp = \frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t} \qquad \tau_\perp = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$$

$$1 + \Gamma_\perp = \tau_\perp$$

$\theta_{B\perp} = \tan^{-1}\left(\sqrt{\frac{\mu_2}{\mu_1}}\right) \longrightarrow$ Practically difficult

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$\theta_c = \{\theta_i : \theta_t = 90^\circ\} \Rightarrow \theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$ (non-magnetic) $(n_1 > n_2)$

at $\theta_i = \theta_c$; $\Gamma_\perp = 1$

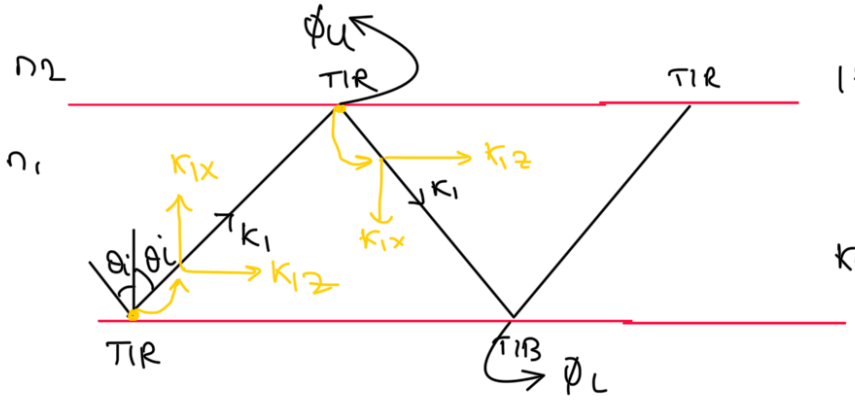
If $\theta_i > \theta_c \Rightarrow \Gamma_\perp = \frac{a - jb}{a + jb}$ $\quad a = n_1 \cos\theta_i$

$b/n_2 = \sqrt{\left(\frac{n_1}{n_2}\right)^2 \sin^2\theta_i - 1}$

$\Rightarrow \Gamma_\perp = e^{-j\phi}$ $\quad \phi = 2\tan^{-1}\left(\frac{b}{a}\right) = \frac{E_{r,0}}{E_{i,0}}$

$\Rightarrow E_{r,0}$ is exactly equal to $E_{i,0}$ but with a phase → Total Internal Reflection

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$E = E_0 e^{i(\omega t \pm K_x x - K_z z)}$

$= E_0 (e^{+iK_x x} + e^{-iK_x x}) e^{i(\omega t - \beta z)}$ $\quad (\beta = K_z)$

$= (2E_0 \cos K_x x)\, e^{i(\omega t - K_z z)}$

Round trip phase acquired: $2K_x \cdot H - \phi_U - \phi_L = 2m\pi$

$$\phi_u = \phi_L = 2\tan^{-1} \frac{n_1 \cos\theta_i}{\sqrt{\left(\frac{n_2}{n_1}\right)^2 \sin^2\theta_i - 1}}$$